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Quantitation of intact parabens in human breast milk from lactating women via fabric-phase sorptive extraction coupled with LC–ESI–MS/MS

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ABSTRACT

Human breast milk is a complex and dynamic biofluid that provides complete nutrition and supports both immune and neurodevelopmental functions in infants. However, its composition may promote the accumulation of environmental contaminants such as parabens. This study introduces an LC–ESI–MS/MS method for the simultaneous quantitation of methylparaben, ethylparaben, propylparaben, and butylparaben in human breast milk. Sample preparation was carried out using fabric phase sorptive extraction (FPSE), with the extraction conditions systematically optimized through a design of experiments (DoE) approach based on a fractional factorial design. Key parameters, extraction volume, extraction time, desorption time, membrane type, and solvent composition, were evaluated to maximize analyte recovery. High performance liquid chromatography was implemented within 6 min on a Nucleodur® pentafluorophenylpropyl analytical column (125 × 2.0 mm, 5 μm) under isocratic conditions. The method employed negative electrospray ionization, using two transitions per analyte for simultaneous quantitation and confirmation. Under optimized conditions, the method achieved recoveries ranging from 76.7 to 106.1 %, with minimal matrix effects between −11.4 and − 4.8 %. Linearity was observed across the concentration range of 1 to 200 ng mL^{-1} for all analytes. The method was applied to breast milk samples from 33 lactating women, revealing the presence of intact parabens in a substantial proportion of samples, with methylparaben detected in 72.7 % of cases. This study is the first to employ FPSE for paraben analysis in breast milk, offering a sensitive and efficient tool for monitoring infant exposure to these compounds.

1. Introduction

In the early stages of life, human breast milk functions not only as a complete source of nutrition but also as a critical foundation for immune protection and neurodevelopment [1]. It is a dynamic and complex biofluid, comprising more than 200 identified components, including macronutrients, micronutrients, and a range of non-nutritive bioactive compounds [2]. The biochemical composition of breast milk changes markedly throughout lactation. Fat content increases over time, from approximately 3–4 % in early milk to over 6 % in mature milk, while pH typically ranges from 7.0 to 7.4 [3]. The transfer of chemicals into breast milk is governed by various factors, including molecular weight, lipophilicity, plasma protein binding, ionization, and pKa [4]. Moderately lipophilic and lipophilic compounds, especially those with low plasma protein binding and moderate molecular weight, tend to diffuse more readily across the mammary epithelium into breast milk [5]. Among

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these, parabens have raised concern due to their widespread use and potential endocrine-disrupting effects [6,7].

Parabens, including methylparaben (MPB), ethylparaben (EPB), propylparaben (PPB), and butylparaben (BPB), are widely used as preservatives in cosmetics, pharmaceuticals, and food products to prevent microbial growth and extend shelf life [8]. Their ability to mimic estrogen has raised safety concerns, particularly for long-chain parabens like PPB and BPB, which exhibit greater estrogenic activity [9]. The estrogenic potency of parabens has been shown to correlate with the length and structure of their alkyl chains, increasing with longer linear chains, branching, or the addition of aromatic substituents such as a benzyl group [10]. Although p-hydroxybenzoic acid, the primary metabolite of parabens, has considerably lower estrogenic activity than the parent compounds, measurable levels of intact paraben esters have been detected in various human biological matrices [11]. The proportion of intact parabens excreted in urine appears to vary according to the route of exposure [12].

In the European Union, regulatory authorities have assessed the safety of parabens and established concentration limits for their use in cosmetic products [13–15]. These evaluations have found that certain parabens are considered safe when used under defined conditions and within specific concentration thresholds. However, targeted assessments of exposure in infants and young children have led to more stringent restrictions for cosmetics intended for this age group [16]. Since 2014, several parabens, including isopropylparaben, isobutylparaben, benzylparaben, phenylparaben, and pentylparaben, have been completely banned from cosmetics products [17]. Despite these measures, the continued and widespread use of parabens in consumer goods underscores the need for regular monitoring and strict compliance with regulatory guidelines, particularly to safeguard vulnerable populations such as breastfed infants and toddlers [18,19].

Several biomonitoring methods for parabens have primarily focused on matrices such as urine, blood, tissues, and hair, leaving a critical gap in our understanding of exposure during lactation [20–22]. Analyzing parabens in breast milk poses considerable challenges due to the matrix's complexity, including high fat content and various endogenous compounds that can interfere with detection [23]. To overcome these obstacles, researchers have explored a range of sample preparation and detection techniques. These include on-line solid-phase extraction (SPE) coupled with LC–MS/MS in APPI mode for phenols and parabens [24], microdisk SPE with LC–UV [25], and protein precipitation followed by LC–ESI–MS/MS for defatted milk [26,27]. Stir-membrane-based solid-liquid-liquid microextraction (SM-SLLME) [28,29], ultrasound-assisted extraction UAE-SLLME [30], in-tube solid phase micro extraction (SPME) [31], solid phase extraction (SPE) [32,33], and liquid–liquid extraction (LLE) [34–36] combined with mass spectrometric detection have also been reported. Other approaches have used QuEChERS following enzymatic hydrolysis [37] or QuEChERS fatty dispersive-solid phase extraction, which requires large sample volumes [38,39] and is coupled with mass spectrometric detection. Notably, one study using a magnetically assisted matrix solid phase dispersion (MA–MSPD) followed by a dispersive liquid–liquid microextraction (DLLME) method combined with LC–ESI–MS/MS reported paraben concentrations in breast milk as high as 3270 ng mL^{-1}, considerably exceeding the typical literature values, which generally fall below 50 ng mL^{-1} for individual parabens [40]. Additional techniques include magnetic solid-phase extraction with Fe_3O_4/graphene oxide/DEHPA coupled with LC–UV [41], a stir bar sorptive extraction (SBSE) [42], and a recent non-targeted LC–TOF–MS screening that detected both regulated and emerging paraben-related contaminants in breast milk [43].

Fabric-phase sorptive extraction (FPSE) is a novel, environmentally friendly sample preparation technique that provides a rapid, cost-effective alternative to traditional methods [44,45]. Its high sorptive capacity, minimal solvent usage, and ability to directly recover analytes make it especially suitable for complex, lipid-rich biological fluids like breast milk. To our knowledge, no existing analytical method has utilized FPSE for the quantitation of intact parabens in human breast milk, a challenging matrix that demands both high sensitivity and selectivity. This study fills this gap by developing and validating a new FPSE-based approach for the direct extraction of target analytes from breast milk without the need for extensive pretreatment or hydrolysis. The objective of this study was to develop and validate a novel bioanalytical method for the simultaneous quantitation of selected intact parabens in human breast milk, employing FPSE combined with liquid chromatography–tandem mass spectrometry (LC–MS/MS). A Design of Experiments (DoE) approach was used to optimize critical parameters in the FPSE process. The validated method was then applied to analyze 33 breast milk samples from lactating women in Greece and Cyprus, as the first FPSE-based study of intact parabens in breast milk for this region. This application not only demonstrates the method's robustness and adaptability but also offers valuable new data on paraben exposure in postpartum women from two European populations.

2. Materials and methods

2.1. Chemicals and reagents

All LC–MS grade organic solvents were sourced from E. Merck (Darmstadt, Germany). Formic acid of analytical purity was also obtained from E. Merck. High-purity HPLC-grade water was prepared using a Direct-Pure Water System, RO 10 (Rephile Bioscience Inc., Europe), which produces reverse osmosis (RO) water from tap water. This was further purified to ultrapure (Type 1) water using a Millipore Synergy® UV water purification unit (Millipore, MA, USA), featuring a 185 nm photooxidation UV lamp for the removal of organic contaminants. MPB, EPB, PPB, and BPB standards were purchased from Acros Organics (New Jersey, USA).

2.2. Sample collection

Pooled blank human breast milk for method validation and breast milk samples from healthy volunteers were obtained from the General Children's Hospital "Panagiotis & Aglaia Kyriakou" in Athens, Greece. The study protocol received approval from the hospital's Ethics Committee (Approval No. 13028/21.07.2021). Written informed consent was obtained from all participants before sample collection. All samples were anonymized, coded and stored at −20 °C, in low-binding polypropylene Eppendorf tubes, until analysis via LC–MS/MS.

2.3. Instrumentation

LC–MS/MS was used to quantify the selected parabens in human breast milk. The system consisted of an AB SCIEX QTRAP 6500+ triple quadrupole mass spectrometer equipped with an IonDrive™ Turbo V electrospray ionization (ESI) source, coupled to a SCIEX ExionLC AD system, which included a binary gradient pump, autosampler, column oven, and system controller. Data acquisition and processing were conducted using Analyst® version 1.7.3 and SCIEX OS® version 3.0 software. The mass spectrometer operated in negative electrospray ionization (ESI−) mode, which was selected for optimal sensitivity for all target analytes. The curtain gas was set to 35 psi, the collision gas was set to high, and both the nebulizer gas (GS1) and drying gas (GS2) were maintained at 40 psi. The source temperature and spray voltage were 250 °C and − 4500 V, respectively. Chromatographic separation was performed on a Nucleodur® pentafluorophenylpropyl (PFP) analytical column (125 × 2.0 mm, 5 μm particle size, 110 Å; Macherey–Nagel, Germany) under isocratic conditions. The mobile phase consisted of acetonitrile (phase A) and distilled water (phase B), each containing 0.1 % formic acid, mixed in a 50:50 (*v*/v) ratio. The flow rate was set to 0.2 mL min^{-1}, the injection volume was 10 μL, and the column oven temperature was maintained at 27 °C.

Sample mixing and extraction were performed using a Transonic

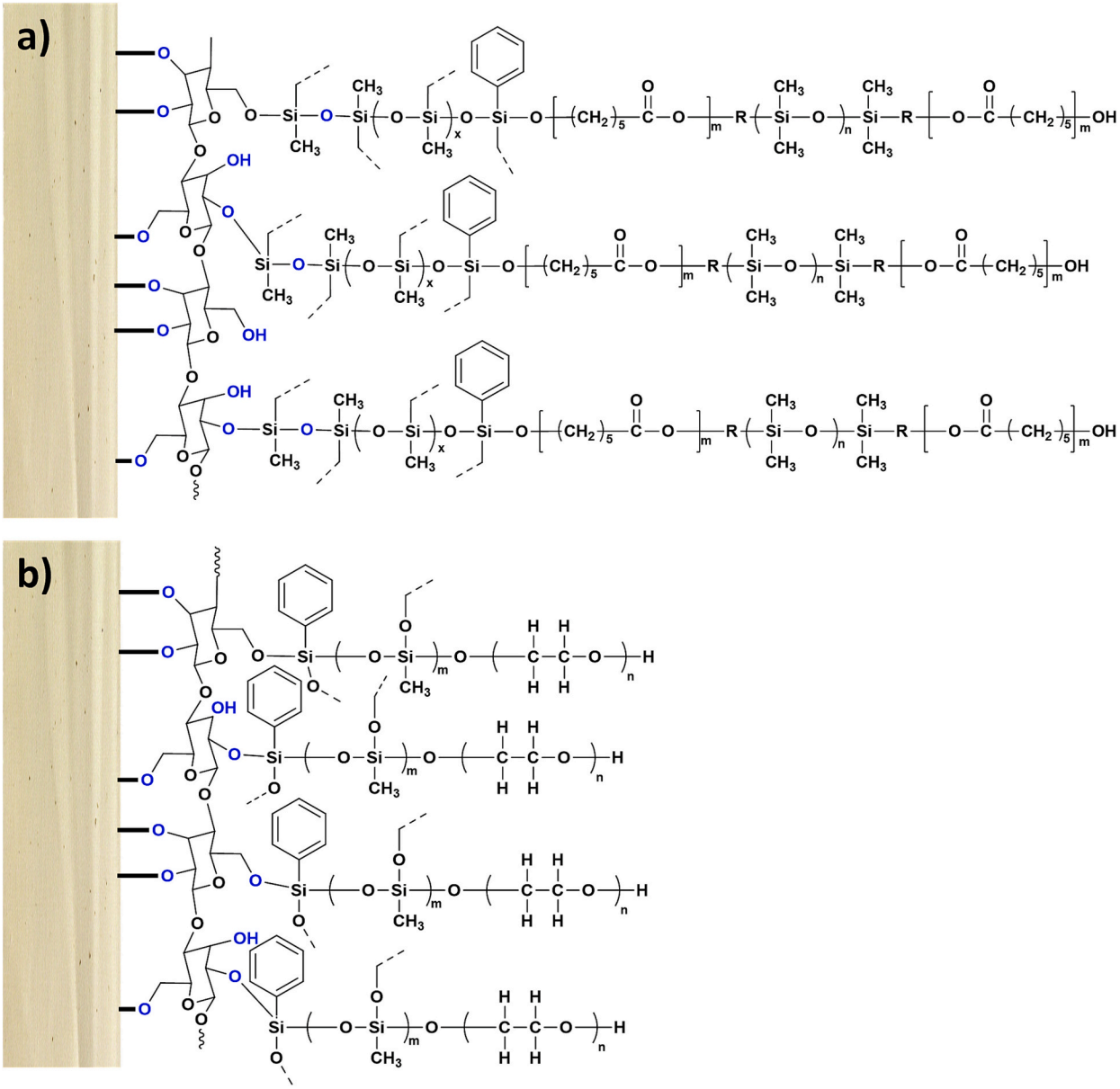


Fig. 1. a): Schematic representation of a) sol-gel methyl-phenyl-poly(caprolactone)-poly(dimethylsiloxane)-caprolactone)-poly(caprolactone) coated FPSE membrane and **b)** sol-gel methyl-phenyl-Carbowax 20 M coating on cellulose fabric.

water bath model 460/H (Elma, Germany) and a vortex mixer (FALC Instruments, Treviglio, Italy), respectively. Magnetic stirring was carried out using Hei-Connect stirrers (Heidolph Instruments, Schwabach, Germany), which provide gentle stirring (up to 110 g) and temperature-controlled heating. The FPSE membranes coated with sol–gel Carbowax® 20 M (sol–gel CW 20 M) and sol–gel methyl-phenyl-poly (caprolactone)-poly(dimethylsiloxane)-poly(caprolactone) [sol–gel PCAP-PDMS-PCAP] used in this study were patented by Kabir and Furton in 2014, and their synthesis procedure is reported elsewhere [46]. Schematic illustrations of the sol–gel CW 20 M and sol–gel PCAP-PDMS-PCAP membranes are presented in Fig. 1a and Fig. 1b, respectively. The geometric dimensions of the FPSE membranes were tailored to meet specific method requirements by using a handheld punch to reproducibly produce either circular disks with a 1.5 cm diameter (surface area: 1.77 cm^2) or square shapes with 1 cm sides (surface area: 1 cm^2).

Hydrophobic polytetrafluoroethylene (PTFE) syringe filters (0.22 μm pore size, 13 mm diameter) used for sample filtration were supplied by Novalab (Athens, Greece). Method optimization through DoE, including fractional factorial design and response surface analysis, was conducted using Design-Expert® software version 23.1.8.0 (Stat-Ease Inc., USA).

2.4. Stock standard solutions and breast milk samples

Individual stock standard solutions of MPB, EPB, PPB, and BPB were prepared in methanol at a concentration of 500 μg mL^{-1}. Mixed working standard solutions were then prepared by appropriate dilution of the stock solutions in water. All stock and working solutions were stored at 8 °C and remained stable for several months.

Matrix-matched calibration standards were prepared by spiking pooled human breast milk with the mixed working solutions, covering a concentration range of 1 to 200 ng mL^{-1} for all analytes. Quality control (QC) samples were independently prepared from separate mixed stock solutions at three concentration levels 1, 40, and 140 ng mL^{-1}, for all target analytes.

Table 1
Retention times (tR) and multiple reaction monitoring conditions for the analysis of intact parabens in human breast milk.

Compound	tR (min)	Precursor ion (Q1, *m/z*)	Product ions (Q3, *m/z*)[1]	Dwell Time (msec)	DP[2]	EP[2]	CE[2]	CXP[2]
Methylparaben (MPB)			**92.1**		−65	−10	−30	−3
	2.78	150.8	135.7	150	−65	−10	−20	−13
Ethylparaben			**135.7**		−30	−10	−22	−21
(EPB)	3.25	164.8	136.5	150	−30	−10	−22	−19
Propylparaben (PPB)			**92.2**		−5	−10	−26	−55
	3.94	179.0	135.6	150	−5	−10	−38	−55
Butylparaben			**91.9**		−5	−10	−34	−51
(BPB)	4.91	192.9	135.9	150	−5	−10	−22	−15

Source parameters: curtain gas (CUR), 35 psi; collision gas (CAD), set to "high"; ion source gas 1 (GS1) and gas 2 (GS2), 40 psi each; source temperature, 250 °C; spray voltage, −4500 V.
[1] Quantitation ion are shown in bold.
[2] DP (Declustering Potential), EP (Entrance Potential), CE (Collision Energy), and CXP (Collision Cell Exit Potential) are reported in volts (V).

2.5. Sample preparation

After thawing at room temperature, the homogeneity of the human breast milk samples was ensured by vortex mixing before sample preparation. No defatting step was performed prior to FPSE, as the membrane allows direct extraction from the aqueous-diluted breast milk matrix. The sol–gel Carbowax® 20 M FPSE membrane was cut into circular disks with a diameter of 1.5 cm (surface area: 1.77 cm^2) and handled using laboratory tweezers to prevent contamination.

The FPSE protocol involved four distinct steps, each performed in separate screw-cap glass vials. First, the sol–gel Carbowax® 20 M FPSE membrane was activated by immersing it in 2 mL of an acetonitrile/methanol mixture (50:50, *v*/v) for 5 min. Next, the membrane was rinsed in 2 mL of deionized water for 5 min to remove residual organic solvents. After activation, the membrane was transferred to vials containing 50 μL of human breast milk sample, 100 μL of 2.5 % (*v*/v) aqueous formic acid, or 850 μL of deionized water. The extraction was then performed under magnetic stirring at 25 *g* for 30 min at room temperature. Finally, the membrane was immersed in 500 μL of acetonitrile for 5 min at room temperature without stirring to allow desorption. The resulting extract was then filtered through a hydrophilic PTFE syringe filter (0.45 μm pore size) and transferred to LC–MS/MS vials for analysis. Because acetonitrile serves as both the desorption solvent and sample diluent, no evaporation or reconstitution steps were required, considerably simplifying the protocol.

2.6. Method validation

The linearity of the LC–ESI–MS/MS method was assessed over four independent analytical runs using matrix-matched calibration standards over the concentration ranges provided in Section 2.4. Each calibration point was analyzed in duplicate, and eight points calibration curves were generated by plotting the analyte peak area against nominal concentration. Linearity was considered acceptable with a correlation coefficient, r, of 0.98 or higher.

Method trueness and precision were assessed by analyzing QC samples at three concentration levels of 1, 40, and 120 ng mL^{-1}. Each level was tested in five replicates per run over three separate analytical runs. Intra-assay (repeatability) and inter-assay (between-run) precision were expressed as the coefficient of variation (CV %). Trueness was calculated by comparing the measured QC concentrations to their nominal values and expressed as percentage recoveries. Following standard validation criteria, trueness and precision were considered acceptable when they were within ±15 % of the nominal concentration, except at the lower limit of quantitation (LOQ), where deviations up to ±20 % were allowed.

Extraction recovery (RE%) was determined by comparing the mean peak areas of QC samples spiked before extraction (pre-spiked) with those of blank breast milk samples spiked after extraction (post-spiked).

Matrix effects were evaluated by analyzing post-spiked samples at two concentration levels, 1 and 40 ng mL^{-1}, and comparing the analyte responses to those of standard solutions prepared in acetonitrile at equivalent concentrations. The matrix effect (M.E.%) was calculated using the following equation:

$$\text{M.E.}(\%) = \left(\frac{B}{A} - 1\right) \text{x } 100 \quad (1)$$

where A is the peak area of each analyte in the neat standard solution, and B is the peak area in the post-extraction matrix-spiked sample. An M.E. close to 0 % indicates no matrix effect, an M.E. > 0 % denotes ion enhancement, and an M.E. < 0 % indicates ion suppression.

Real samples were quantified using matrix-matched calibration and QC samples to account for any variability in MS response.

3. Results and discussion

3.1. Method development

3.1.1. LC–ESI–MS/MS optimization

Mass spectrometric parameters were individually optimized for each analyte to maximize sensitivity in negative electrospray ionization (ESI−) mode. Compound-specific transitions (Q1 and Q3), along with parameters such as declustering potential (DP), collision energy (CE), and collision cell exit potential (CXP), were fine-tuned via direct infusion of standard solutions at 10 ng mL^{-1} for each analyte. For subsequent quantitation, the most intense transition for each analyte was selected as the quantifier ion (see Table 1). The selected product ions follow paraben fragmentation pathways including initial loss of the alkyl ester moiety, yielding $m/z \sim 136$, and neutral loss of CO_2, yielding $m/z \sim 92$, as described in previous LC–ESI–MS/MS studies [47].

During chromatographic method development, several columns were evaluated, including Kinetex EVO C18, ZIC-pHILIC, and Nucleodur® pentafluorophenylpropyl (PFP). The PFP column was ultimately chosen for its superior retention, low background signal, and excellent compatibility with LC–MS/MS detection. This stationary phase, made from high-purity Nucleodur silica gel functionalized with pentafluorophenylpropyl ligands, offers enhanced selectivity through combined hydrophobic, π–π, dipole–dipole, and hydrogen-bonding interactions. Its low bleed profile further ensures robustness and reproducibility in LC–MS applications.

Various mobile phase compositions were evaluated using acetonitrile concentrations from 40 to 70 % in aqueous ammonium formate at 5 and 10 mM. Although 45 % acetonitrile in 5 mM ammonium formate achieved acceptable retention for all analytes, replacing the buffer with 0.1 % formic acid considerably enhanced the peak symmetry and signal intensity. The final optimized conditions employed isocratic elution with 50 % acetonitrile and 50 % water, both containing 0.1 % formic acid, at a flow rate of 0.2 mL min^{-1}, with the column oven set to 27 °C.

Table 2
Regression equations in terms of coded factors and statistical fit parameters for process efficiency (%PE) of each paraben.

Response- %Process E=efficiency[1]	Transformation	Regression equation (coded factors)[2]	Pred. R^2	Adj. R^2	Significant terms ($p < 0.05$)
PE_{MPB}	Natural Log	$\ln(PE_{MPB}) = 2.47–0.58\ A + 0.17B - 0.98E - 0.13F + 0.04AB - 0.05AE - 0.02BE - 0.06ABE$	0.99	0.991	A, B, E, F, AE, ABE
PE_{EPB}	Square Root	$\sqrt{(PE_{EPB})} = 5.69–1.11\ A + 0.59B - 2.01E - 0.25F + 0.18AB + 0.08AE - 0.16BE - 0.25ABE$	0.97	0.98	A, B, E, F, AB, BE, ABE
PE_{PPB}	Square Root	$\sqrt{(PE_{PPB})} = 5.82–1.15\ A + 0.61B - 2.05E - 0.26F + 0.18AB + 0.09AE - 0.17BE - 0.12EF - 0.26ABE$	0.97	0.98	A, B, E, F, AB, BE, EF, ABE
PE_{BPB}	None	$PE_{BPB} = 46.27–13.03\ A + 8.76B + 2.18D - 22.70E - 3.23F + 2.49AB + 2.71AE + 1.92AF - 3.84BE - 3.79ABE$	0.96	0.98	A, B, D, E, F, AE, AF, BE, ABE

[1] PE_{MPB}, PE_{PPB}, PE_{BPB}, and PE_{EPB}: % process efficiency of methylparaben, propylparaben, butylparaben and ethylparaben, respectively.
[2] Coded factors: A = Extraction volume (mL), B = Extraction time (min), C=Desorption time (mL), D=Desorption solvent, E = FPSE membrane type, and F = FPSE membrane shape.

Under these conditions, chromatographic separation was achieved in 6 min, with retention times for MPB, EPB, PPB, and BPB recorded at 2.83, 3.30, 4.01, and 4.98 min, respectively, following analysis of matrix-matched calibration samples.

3.1.2. Optimization of the FPSE protocol

FPSE is a novel technique that combines key features from both SPE and SPME. Unlike conventional methods, FPSE employs a porous, fabric-based substrate coated with a sol–gel-derived sorbent, allowing it to function through both exhaustive and equilibrium-driven mechanisms. When immersed directly into an aqueous sample, the membrane behaves similarly to SPME, engaging in passive, equilibrium-based extraction. However, due to the intrinsic porosity of the fabric support, even after sol–gel coating, the sample solution repeatedly passes through the membrane structure during stirring, effectively mimicking the flow-through mode of SPE. This continuous circulation across the sorbent layer enhances analyte-sorbent interaction and leads to high extraction efficiency. In contrast to conventional SPE, where the sample flows through a disk or cartridge just once under pressure, FPSE allows for spontaneous and repeated permeation without the need for pumps or vacuum. This dual mechanism of action supports reliable and sensitive analyte recovery, even in complex biological matrices such as breast milk, making FPSE a robust platform suitable for both laboratory and field applications.

Previous work on FPSE for the analysis of intact parabens in human plasma provided valuable insights into the influence of critical parameters [48]. Based on previous findings, a sol–gel Carbowax® 20 M coated FPSE membrane, known for its medium polarity, was selected for its affinity toward moderately polar compounds such as parabens. Simultaneously, a sol–gel PCAP–PDMS–PCAP membrane, representing a more hydrophobic sorbent, was also evaluated. To improve the retention of weakly acidic parabens, whose pKa is around 8.5, the extraction was carried out under mildly acidic conditions to favor their neutral form. Based on this findings and preliminary optimization trials (data not shown), the method was developed using a 50 μL aliquot of breast milk, 0.25 % formic acid in water as the extraction solvent, continuous

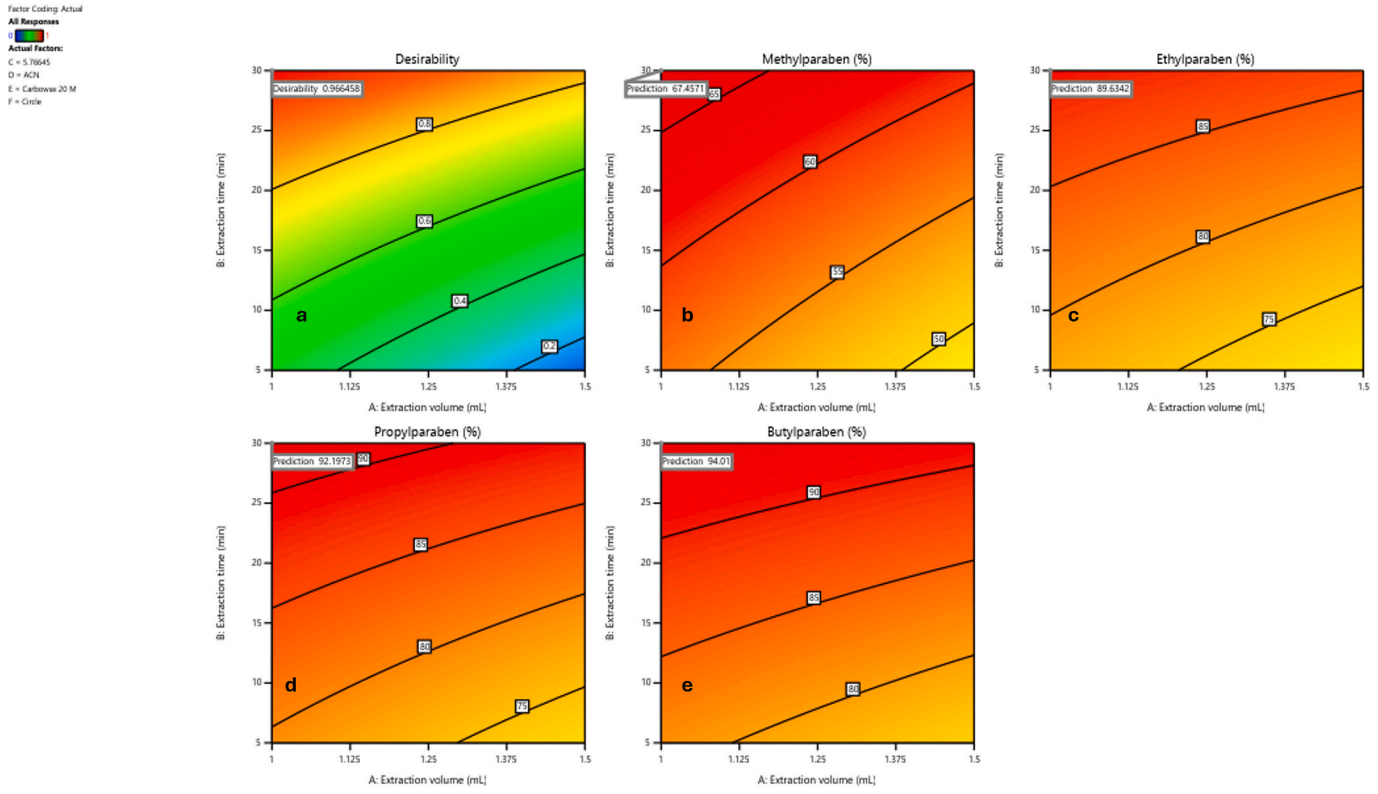


Fig. 2. Contour plots depicting (a) the overall desirability score and the influence of extraction volume (A) and extraction time (B) on process efficiency for (b) methylparaben, (c) ethylparaben, (d) propylparaben, and (e) butylparaben.

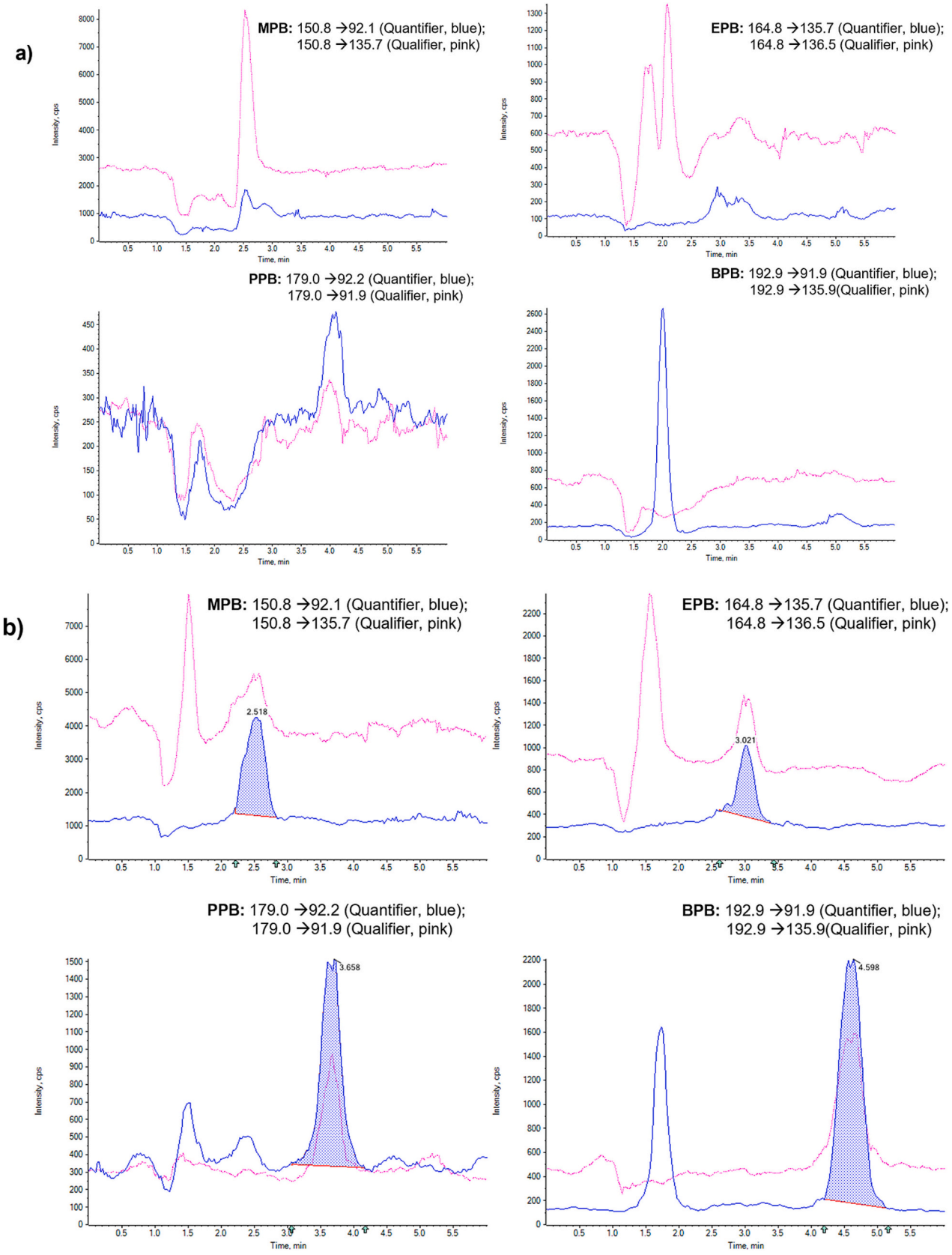


Fig. 3. Extracted ion chromatograms (XICs) of methylparaben (MPB), ethylparaben (EPB), propylparaben (PPB), and butylparaben (BPB) in (a) a blank human breast milk sample and (b) a matrix-matched breast milk sample spiked at 1 ng/ mL. The blue trace corresponds to the quantifier ion transition, and the pink trace corresponds to the qualifier ion transition for each analyte. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

stirring during extraction, and a 0.5 mL desorption volume.

A DoE approach was employed to examine the impact of various experimental parameters on extraction performance and to optimize the process efficiency for the analytes. Specifically, the effects of extraction and desorption variables on the process efficiency (%) of MPB, EPB, PPB, and BPB were assessed using a 2^{6-1} fractional factorial design. A reduced three-factor interaction (3FI) model was chosen to evaluate the main effects and selected interactions. The design details are provided in Supplementary Table S1. Experiments were conducted using matrix-matched breast milk samples spiked with the target analytes at 200 ng mL^{-1}. Process efficiency (%) was calculated by comparing the peak area signals obtained after extraction to those from standard solutions at equivalent concentrations.

Pareto charts were generated for each analyte to identify the key factors influencing process efficiency (%), as shown in Figs. S1a to S1d. These charts illustrate the standardized effects of all main factors and their interactions. Factors with effects exceeding the Bonferroni and *t*-test thresholds guided the model selection process. For all analytes, membrane type (E) and extraction volume (A) exhibited the strongest negative effects, whereas extraction time (B) consistently had a positive impact on process efficiency. Although membrane shape (F) showed a less pronounced effect, it remained significant across all models. Table 2 presents the final regression models in coded form, including their statistical parameters and significant factors identified via ANOVA. Each model demonstrated robust performance, with predicted R^2 values between 0.96 and 0.98 and no significant lack of fit.

Response surface plots were generated under optimized conditions to illustrate the effects of extraction volume and time on each analyte (Figs. 2b to 2d). These plots confirm that maintaining extraction volume below 1.25 mL enhanced the process efficiency across all parabens, with notably strong effects for PPB and BPB. To determine the optimal conditions for maximizing process efficiency across all analytes, we conducted a numerical optimization using Derringer's desirability function (Fig. 2a). As indicated in Table 2, the type of desorption solvent (D) was a significant factor only for BPB, where methanol provided a slight improvement in process efficiency. In contrast, no statistically significant effect of the desorption solvent was found for MPB, EPB, or PPB. Therefore, acetonitrile was chosen as the desorption solvent because it also was used as the sample diluent for LC–MS/MS analysis, eliminating the need for evaporation and reconstitution steps. Overall, the optimal FPSE conditions, with a desirability score of 0.97, were: an extraction volume of 1.0 mL, extraction time of 30 min, sol–gel CW 20 M membrane cut into a circular disk, and desorption time of 5.5 min. The optimized FPSE protocol is described in detail in Section 2.5.1.

3.2. Method validation

3.2.1. Selectivity and carryover

To confirm selectivity, blank samples from six different donors were analyzed using the optimized sample preparation procedure, and no interfering peaks exceeding 20 % of the LOQ were detected at the retention times of the target analytes. Fig. 3 presents XIC chromatograms from (a) a blank breast milk sample and (b) matrix-matched samples spiked at 1 ng mL^{-1} (LOQ). Consistent with the acceptance criteria outlined in the ICH M10 guidelines for bioanalytical method validation, no interfering peaks surpassing 20 % of the analyte response at the LOQ were observed in the blank samples at the expected retention times [49]. To assess carryover, blank breast milk samples were analyzed immediately after the highest QC level, revealing no peaks above 20 % of the LOQ at the analytes' retention times, indicating minimal carryover.

Sensitivity was assessed using signal-to-noise (S/N) ratios, with limits of detection (LOD) and quantitation (LOQ) calculated as LOD = 3 × (S/N) and LOQ = 10 × (S/N), respectively. The signal was determined from the peak area at the lowest validated concentration of 1 ng mL^{-1}, while noise was measured in a defined region surrounding the analyte peak in a blank sample [46]. Using this method, LOD values of 0.2 ng mL^{-1} and LOQ values of 1 ng mL^{-1} were obtained for all analytes.

Table 3
Method validation data.

Weighted linear regression parameters (mean ± SD, $n = 4$)[1].

Analyte	Slope	y-intercept	Correlation coefficient, r	Standard error of estimate, Sr
Methylparaben	23,179 ± 2409	8959 ± 7690	0.998 ± 0.001	0.1 ± 0.1
Ethylparaben	8553 ± 716	2714 ± 1731	0.9998 ± 0.0002	13.8 ± 6.5
Propylparaben	16,775 ± 1096	4966 ± 4425	0.9998 ± 0.0001	16.1 ± 6.5
Butylparaben	30,905 ± 1715	12,091 ± 8819	0.99991 ± 0.00006	22.8 ± 3.1

Precision and trueness data

Analyte	Spiked ng mL^{-1}	Overall mean	Intraday CV(%)[2]	Total precision CV (%)[2]	Total trueness % Recovery[2]
MPB	1	1.01 ± 0.08	8.4	10.5	100.2
	40	42.68 ± 2.17	5.6	5.8	106.7
	120	128.32 ± 5.80	4.9	5.4	106.9
EPB	1	1.01 ± 0.06	5.5	8.4	100.1
	40	42.29 ± 1.93	4.6	5.9	105.7
	120	129.51 ± 4.11	2.6	5.6	107.9
PPB	1	1.06 ± 0.06	5.2	5.9	106.4
	40	41.64 ± 1.52	2.9	6.7	104.1
	120	130.52 ± 4.16	3.0	5.0	108.8
BPB	1	0.99 ± 0.06	5.5	8.4	99.5
	40	41.03 ± 0.84	2.2	2.4	102.6
	120	128.99 ± 3.96	3.3	3.8	107.5

[1] Weighted linear regression was applied using a $1/Y^2$ weighting factor for methylparaben, and a 1/Y weighting factor for ethylparaben, propylparaben, and butylparaben (BPB).

[2] Coefficient of variation; intra- and inter-assay CVs were calculated by one-way ANOVA (SPSS v29.0.2.0).

3.2.2. Linearity

The linearity of the proposed method was assessed by constructing four independent calibration curves over a 1-month period. Least squares regression with a weighting factor of $1/Y^2$ was applied for MPB, while a 1/Y weighting was used for EPB, PPB, and BPB. The results of the weighted linear regression analysis are summarized in Table 3, which presents the mean values and standard deviations (SD) of slopes, y-intercepts, and correlation coefficients (r) from the calibration curves. Correlation coefficients exceeded 0.991 for all analytes. At the LOQ level, back-calculated concentrations deviated by no more than 18.2 %, with deviations at all other concentration levels remaining within 13.1 %, consistent with ICH M10 guidelines [49].

3.2.3. Trueness and precision

The ion ratios of all analytes in quality control and study samples were within ±20 % of the values established during the analysis of matrix-matched calibration samples. Trueness and precision results are summarized in Table 3. The method demonstrated excellent trueness,

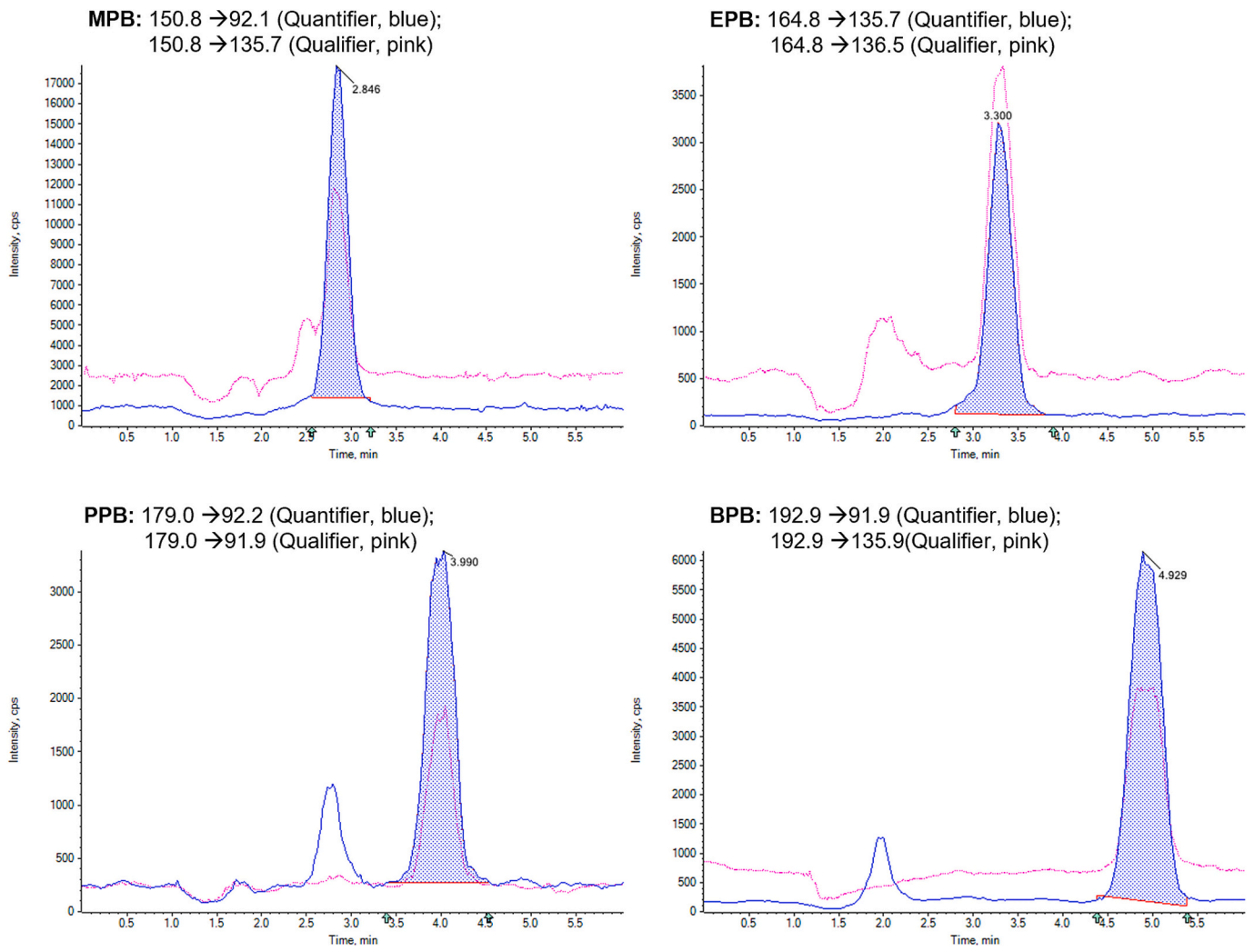


Fig. 4. Extracted ion chromatograms (XICs) of methylparaben (MPB), ethylparaben (EPB), propylparaben (PPB), and butylparaben (BPB) obtained from a real breast milk sample of a 20-year-old lactating mother. The quantified concentrations of MPB, EPB, PPB, and BPB were 7.8, 5.1, 3.6, and 3.9 ng mL^{-1}, respectively.

with recovery values ranging from 99.5 to 108.8 % across all analytes and concentration levels. Intraday precision, expressed as the coefficient of variation (CV), remained below 8.4 %, while total CVs were within 10.5 %, complying with the acceptance criteria outlined in the ICH M10 guidelines [46]. These findings confirm the method's reliability and reproducibility for quantifying analytes in human breast milk.

3.2.4. Matrix effects and recovery

Matrix effect and recovery results are summarized in Table S2. Experimental recoveries ranged from 76.7 to 106.1 %, while matrix effects varied between −11.4 and − 4.8 %, indicating consistently low ion suppression across all analytes. Recovery increased with analyte lipophilicity, with BPB exhibiting the highest recovery, followed by PPB, EPB, and MPB. This trend reflects favorable partitioning into the sol–gel Carbowax® 20 M membrane, which supports both polar and hydrophobic interactions. Predicted and experimental process efficiencies were closely aligned, with deviations below 10 %, and matrix effects within the accepted ±15 % range.

Table 4
Results of the LC-MS/MS analysis of the intact parabens in human breast milk samples collected from 33 women in Greece and Cyprus.

Compound	MPB	EPB	PPB	BPB
Percentage of samples quantified (≥LOQ)	60.6	39.4	21.2	12.1
Percentage of samples detected (>LOD and < LOQ)[1]	12.1	15.2	15.2	21.2
Mean concentration[1] (ng mL^{-1})	5.73	3.51	3.77	5.33
Median concentration[1] (ng mL^{-1})	3.15	3.56	1.76	2.50
Concentration range[1] (ng mL^{-1})	1.0–24.9	1.0–11.8	1.0–9.1	1.1–15.3

[1] Mean, median, and range calculated using only values ≥ LOQ (1.0 ng mL^{-1}); LOD = 0.2 ng mL^{-1}.

3.2.5. Stability

To evaluate analyte stability, human breast milk samples were spiked with 100 ng mL^{-1} of MPB, EPB, PPB and BPB, and stored at −20 °C in low-binding polypropylene Eppendorf tubes. Samples were assessed after four weeks of storage and again following four freeze–thaw cycles, each carried out at weekly intervals. Across all four compounds, the average change in concentration was −5.8 % (± 1.9) after storage and − 6.5 % (± 2.8) following freeze–thaw cycles, indicating that the analytes remained stable under these conditions.

3.3. Application to real sample analysis

All breast milk samples analyzed using the proposed method were collected from women between one and four months postpartum. For all positive samples, the qualifier-to-quantifier ion ratios were within the acceptable ±20 % tolerance range. Representative extracted ion chromatograms for MPB, EPB, PPB, and BPB from a real sample of a 20-year-

Table 5
Comparison of the FPSE-LC-ESI-MS/MS method to other liquid chromatographic methods for the quantitation of the targeted analytes in human breast milk.

Analytes[1]	Technique	LC run time (min)	Sample preparation[2]	Extraction time (min)	Sample volume (mL)	LOD (ng mL^{-1})	Number of real samples	Reported parabens' levels (ng mL^{-1})	Precision (CV%)	Ref.
MPB, EPB, PPB, BPB and other compounds	LC-ESI-MS/MS	5	Defatted milk /PP	20	0.2	1.0	54	MPB 2.18 ± 2.02; EPB 1.26 ± 0.23; PPB 1.42 ± 0.38	Not reported	Schlumpf et al. 2010 [25]
MPB, EPB, BPB	LC-UV	12	PP and MIP-SPE	Not specified	0.1	10.0–20.0 (LOQ)	3	Not Detected	2.6–12	Melo et al. 2013 [24]
MPB, EPB, PPB, BPB	UHPLC-ESI-MS/MS	10	Freeze dried milk, SM-SLLME	45	1	0.1–0.2	10	MPB 0.9–11.3; EPB 0.5–13.2; PPB 0.6–37; BPB 0.6–11.3	1.1–7.7	Rodríguez-Gómez et al. 2014 [27]
MPB, EPB, PPB, BPB, BzPB and other compounds	LC-APPI-MS/MS	15	On-line SPE	20	0.1	0.1–0.4	4 / 10	MPB 0.7–3; PPB 0.3/ MPB 0.5–2.3; PPB 0.1–0.6	3.5–16.3	Ye et al. 2008 [23]/ Hines et al. 2015 [26]
MPB, EPB, PPB, BPB and bisphenol A	UHPLC–ESI-MS/MS	10	Freeze dried milk, SM-SLLME	70	1	0.1–0.2	10	MPB 1.82–16.3; EPB 0.97–18.1; PPB 1.02–12.6/ BPB 1.06–12.1	1.3–11.1	Rodríguez-Gómez et al. 2015 [29]
MPB, EPB, PPB, BPB	UHPLC-ESI-MS/MS	5	In-tube SPME	3	0.2	MPB, EPB, PPB10 (LOQ); BPB 3 (LOQ)	16	MPB 10.8–39.8; EPB 11.5–29.6; BPB 3–8.1	2.4–15	Souza et al. 2016 [30]
MPB, EPB, PPB	LC-ESI-MS/MS	15	MA-MSPD/ DLLME	27–31	0.2	0.5	6	MPB:116–3270; EPB: 51.2–1499; PPB: 100–1067	8.4–11.3	Fotouhi et al. 2017 [39]
MPB, EPB, PPB, BPB, iBPB, BzP	LC-ESI-MS/MS	15	LLE	Not specified	Not specified	0.07–0.26	56	MPB 0.2–16.3; EPB 0.12–2.183 & PPB 0.33–4.59.	3.8–8.3	Fischer et al. 2017 [33]
MPB, EPB, PPB, BPB and UV filters	LC-ESI-MS/MS	12	QuEChERS d-SPE	17	1.0	0.1–0.2	15	MPB: 0.9–21; EPB:0.1; PPB:0.4–5.8	6.0–13	Vela 2018 [2018]
MPB, EPB, PPB, BPB	LC-MS/MS	15	LLE	–	1.0	0.02–0.04	10/260	MPB 0.1–29.6; EPB 1.2–17.8; PPB: 0.1–8.0; BPB: 0.02–1.12	2.3–8.2 %	Park et al. 2016 & 2019 [34,35]
MPB, EPB, PPB, BPB and 3 bisphenols	LC-APCI-MS/MS	12	QuEChERS d-SPE	50	10	0.1 (LOQ)	10 /120	MPB: 0.11–7.0; EPB 0.49–4.05; PPB: 0.13–0.76; BPB 0.17–0.34/ MPB: 0.1–31; EPB: 0.1–5.2; PPB 0.1–6.5; BPB: 0.1–1.1	1–16 %	Dualde et al. 2019, and 2020 [38]
MPB, PPB, phenol and bisphenol A	HPLC-UV	35	Defatted milk /PP	15	2.0	0.0025–0.014	4	MPB: 0.02–0.29 PPB: 0.06–0.08	1.0–7.9	Abdolmohammad-Zadeh et al. 2020 [40]
MPB, EPB, PPB, BPB, BzPB along with triclosan, triclocarban	UHPLC-ESI-MS/MS	13	LLE-SPE	–	1.0	0.001–0.008	62	MPB: 3.05 ± 1.01 EPB: 0.06 ± 0.02 PPB: 1.34 ± 0.62 BPB: 0.003 ± 0.001 BzPB: 0.002 ± 0.001	–	Zhang et al. 2023 [32]
MPB, EPB, PPB, BPB	UHPLC/UV	10	SBSE	84	2.0	0.01	31	MPB: 0.01–6.47; EPB:0.01–141.82; PPB: 0.01–5.45; BPB: 0.01–9.34	3.1–9.5	Ge et al. 2024 [41]
MPB, EPB, PPB, BPB	LC-ESI-MS/MS	6	FPSE	45	0.05	0.2	33	MPB: 1–24.9; EPB: 1–11.8; PPB: 1–9.1; BPB: 1–15.3	2.4–10.5	This work

[1] methylparaben (MPB), ethylparaben (EPB), propyl-paraben (PPB), butylparaben (BPB), isobutylparaben (iBPB), and benzylparaben (BzPB);
[2] Protein precipitation (PP); solid phase extraction (SPE); molecularly imprinted polymers-solid phase extraction (MIP-SPE); solid phase microextraction (SPME); magnetically assisted matrix solid phase dispersion (MA–MSPD); Stir-membrane solid-liquid-liquid microextraction (SM-SLLME); dispersive liquid-liquid microextraction (DLLME); liquid-liquid extraction (LLE); Quick, Easy, Cheap, Effective, Rugged, and Safe (QuEChERS); magnetic solid-phase extraction (MSPE); stir bar sorptive extraction (SBSE); fabric phase sorptive extraction (FPSE).

old lactating mother are shown in Fig. 4. Quantified concentrations of MPB, EPB, PPB, and BPB were 7.85.1, 3.6, and 3.9 ng mL^{-1}, respectively. Parabens were detected in the majority of the human breast milk samples analyzed ($n = 33$). Of these, 13 samples were collected from Greece and 20 from Cyprus. MPB was the most frequently quantified analyte detected in 60.6 % of samples, followed by EPB in 33.3 %, and both PPB and BPB in 12.2 % each. The highest concentrations measured were 24.9 ng mL^{-1} for MPB and 15.3 ng mL^{-1} for BPB. Besides the quantifiable concentrations, low-level detections between the LOD and LOQ were observed in 12.1, 15.1, 15.1, and 21.2 % of samples (n = 33) for MPB, EPB, PPB, and BPB, respectively. Non-detectable results were most common for BPB, occurring in 66.7 % of samples (Table 4). No apparent differences in paraben concentrations were observed between samples from Greece and Cyprus.

Exposure history collected from lactating women included cosmetic use, vitamin supplementation, and prescribed medications. Elevated MPB concentrations were observed in those who reported using body cream and multiple cosmetic products, such as make-up, sunscreen, and facial moisturizers. For example, lactating women who used body cream in combination with other products showed MPB levels as high as 18.0 ng mL^{-1} and 7.8 ng mL^{-1}. However, this trend was not consistent across all cases, no consistent correlation was found between the number or types of products reported and paraben concentrations in breast milk. Variability in intact paraben levels likely reflects individual differences in product usage frequency, formulation, metabolism, or exposure from unreported sources. To confirm these preliminary patterns and establish any causal relationships, a more detailed and standardized biomonitoring study is necessary. A detailed summary of participant exposure and analyte levels is provided in Supplementary Table S3.

3.4. Comparison with other methods

Following the analysis of real samples, the performance of the developed method was evaluated against existing published methods for paraben determination in breast milk. As summarized in Table 5, numerous liquid chromatographic techniques have been developed for determining parabens in human breast milk using a variety of extraction approaches, including solid-phase microextraction, QuEChERS, magnetic dispersive methods, and liquid–liquid extraction. Many of these methods require relatively large sample volumes, lengthy extraction times, or extensive cleanup procedures. In contrast, the FPSE–LC–ESI–MS/MS method developed in this study provides a rapid, environmentally friendly, and efficient alternative, using only 0.05 mL of sample and 45 min of preparation time. It achieves comparable or lower detection limits with acceptable precision (CV% 2.4–10.5), highlighting FPSE's suitability for high-throughput biomonitoring of intact parabens in complex matrices such as breast milk.

The levels of MPB, EPB, and PPB quantified in this study fall within the range of those previously reported (Table 5). Notably, intact BPB, which is frequently undetected in earlier studies, was quantified in selected samples at concentrations up to 15.3 ng mL^{-1}. This observation aligns with findings from Rodríguez-Gómez et al. (2015) [30], Souza et al. (2016) [31], and Ge et al. (2024) [42], who also reported comparable levels of BPB. In contrast, other studies, such as those by Dualde et al. (2020) [39] and Zhang et al. (2023) [33], reported the presence of BPB at lower concentrations, typically below 1 ng mL^{-1}. These findings reinforce the robustness of the proposed method and emphasize the importance of continued biomonitoring using highly sensitive, low-volume extraction techniques such as FPSE.

3.5. Greenness evaluation of the analytical method

Several approaches have been proposed to assess the performance and environmental impact of analytical methods [50]. In this work, the environmental impact of the FPSE–LC–ESI–MS/MS method was assessed using the ComplexMoGAPI software [51]. The tool evaluates the method across several categories, each containing individual factors related to sample handling, reagent use, instrumentation, and environmental efficiency. The software then produces a cumulative score and a visual pictogram. The method demonstrated favorable characteristics in several key factors such as micro-scale sample preparation, mild conditions with no derivatization step, low energy demand, high analytical recovery, and minimal waste generation. According to the ComplexMoGAPI output, the method achieved a greenness score of 78, indicating a moderate-to-high level of environmental sustainability. The final pictogram and score are shown in Fig. S2.

4. Conclusion

This study presents a novel FPSE method coupled with LC–ESI–MS/MS for the quantitation of intact parabens in human breast milk. The method was optimized through a statistically guided DoE approach and demonstrated strong analytical performance, including low LOQs, high extraction recoveries, and minimal matrix effects. The low sample volume requirement (50 μL) and evaporation-free solvent protocol underscore FPSE's advantages as an efficient, green, rapid, and reproducible sample preparation technique. Matrix-matched calibration was used to address matrix effects, ensuring accurate and precise quantitation of parabens in human breast milk samples. Its application to 33 real breast milk samples confirmed the method's robustness and suitability for trace analysis in lipid-rich biological fluids. This analytical approach offers a valuable tool for future studies on paraben exposure and supports the advancement of eco-friendly sample preparation workflows for complex biological matrices.

CRediT authorship contribution statement

Anthi Parla: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Abuzar Kabir:** Writing – review & editing, Writing – original draft, Visualization, Resources, Methodology, Investigation, Conceptualization. **Marios Kostakis:** Writing – review & editing, Resources, Investigation, Conceptualization. **Kenneth G. Furton:** Writing – review & editing, Resources, Investigation, Conceptualization. **Georgios Mavrogeorgos:** Writing – review & editing, Resources, Investigation. **Kassandra Tataropoulou:** Writing – review & editing, Resources, Investigation. **Anastasia Armefti:** Writing – review & editing, Resources, Investigation. **Victoria Samanidou:** Writing – review & editing, Resources, Investigation, Conceptualization. **Nikolaos Thomaidis:** Writing – review & editing, Resources, Investigation, Conceptualization. **Ivana Vinković Vrček:** Writing – review & editing, Resources, Investigation, Conceptualization. **Irene Panderi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.microc.2025.114774.

Data availability

Data will be made available on request.

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